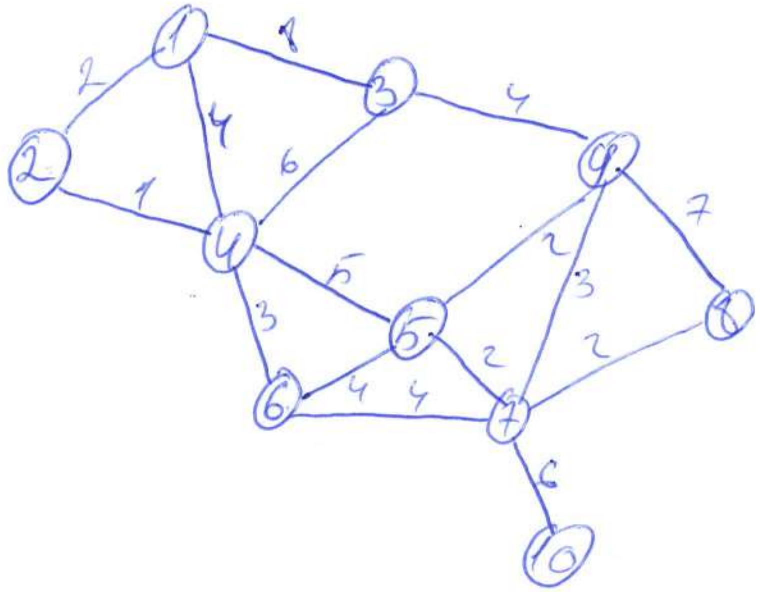
Pavel Marius Florian Adrian
Grupa 222

Laborator 10

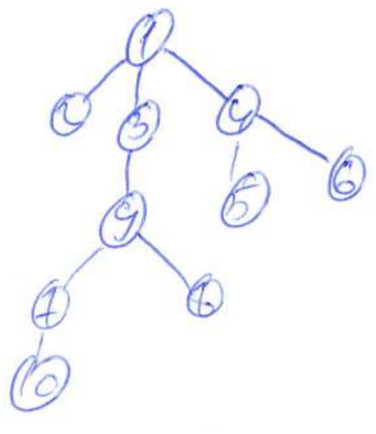
1)



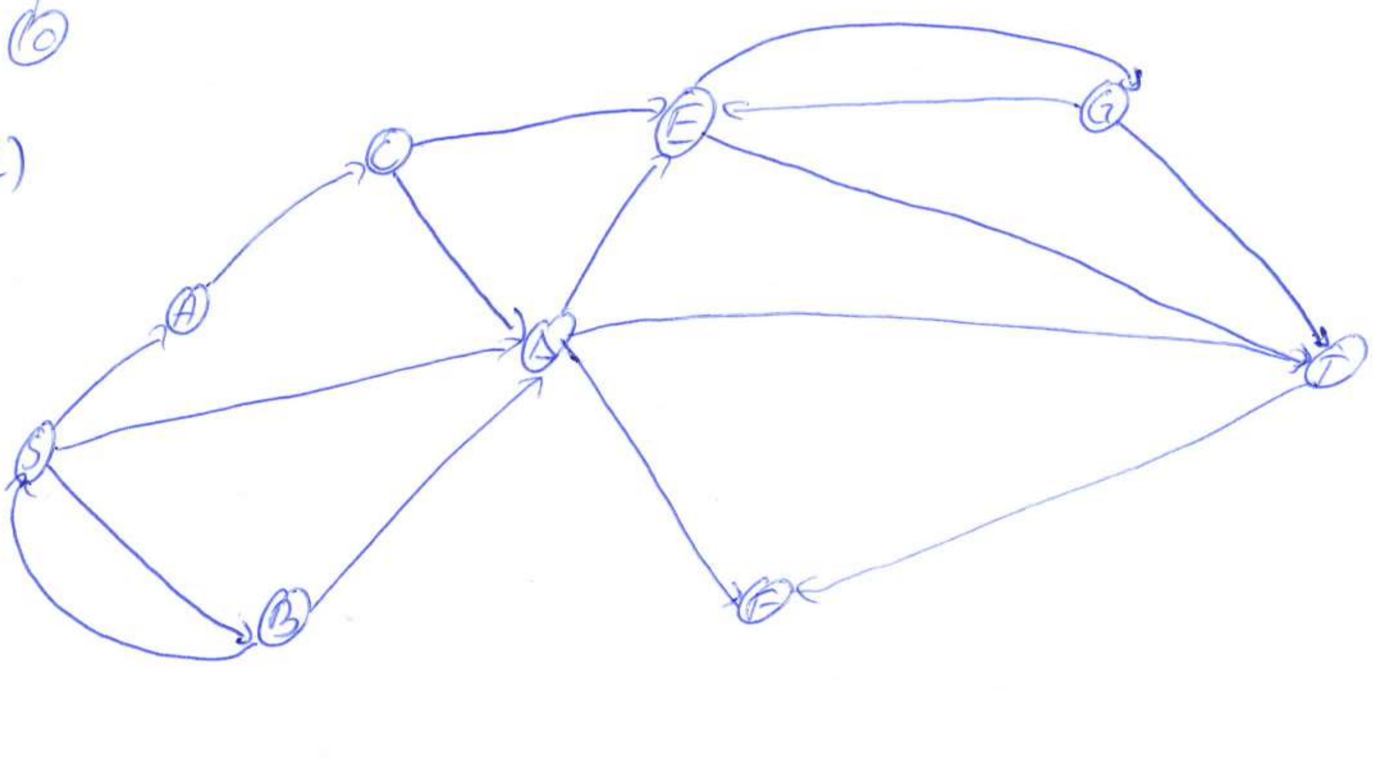
$d(1) = 0$
$d(2) = 0+1 = 1$
$d(3) = 0+1 = 1$
$d(4) = 0+1 = 1$
$d(9) = 1+1 = 2$
$d(5) = 1+1 = 2$
$d(6) = 1+1 = 2$
$d(7) = 2+1 = 3$
$d(8) = 2+1 = 3$
$d(10) = 3+1 = 4$

	1	2	3	4	5	6	7	8	9	10
d	0	∞ =1	∞ =1	∞ =1	∞ =2	∞ =2	∞ =3	∞ =3	∞ =2	∞ =4

Q | ~~1~~ | ~~2~~ | ~~3~~ | ~~4~~ | ~~9~~ | ~~5~~ | ~~6~~ | ~~7~~ | ~~8~~ | 10 |



2)



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2)

	S	A	B	C	D	E	F	G	T
d	0	4	3	~~∞~~ 5	7	~~∞~~ 6	~~∞~~ 12	~~∞~~ 8	~~∞~~ 10
tată	0	S	S	~~8~~ A	S	~~8~~ C	~~8~~ D	~~8~~ E	~~8~~ F
vizitat	1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1	~~0~~ 1

Vecinii lui B, S, D

$d_S > d_B + C_{BS}$? $0 \leq 3+3$? F

$d_D > d_B + C_{BD}$? $7 > 3+4$ $7>7$ F

Vecinii lui A: C

$d_C > d_A + C_{AC}$? $\cancel{\infty} > 4+1$ A

Vecinii lui C: E, D

$d_D > d_C + C_{DC}$ $7 > 5+3$ F

$d_E > d_C + C_{CE}$ $\cancel{\infty} > 5+1$ A
6

Vecinii lui E: G, T

$d_G > d_E + C_{EG}$ $\cancel{\infty} > 6+2$ A
8

$d_T > d_E + C_{ET}$ $\cancel{\infty} > 6+4 = 10$ A

Vecinii lui D: E, F, T

$d_E > d_D + C_{DE}$ $6 > 7+1$ F

$d_F > d_D + C_{DF}$ $\cancel{\infty} > 7+5$ A
12

$d_T > d_D + C_{DT}$ $10 > 7+3$ F

Vecinii lui G: E, T

$d_E > d_G + C_{GE}$ $6 > 8+2$ F

$d_T > d_G + C_{GT}$ $10 > 8+5$ F

Vecinii lui T: F

$d_F > d_T + C_{FT}$ $\cancel{12} > 10+5$ F

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2) F nu are vecini

{A, S} -> {S, A} ~~tabelul~~ tatăl lui A = S

{B, S} -> {S, B} ~~tabelul~~ tatăl lui B = S

{C, A, S} -> {S, A, C} tatăl lui C = A, tatăl lui A = S

{D, S} -> {S, D} tatăl lui D = S

{E, C, A, S} -> {S, A, C, E} tatăl lui E = C A -> S

{F, D, S} -> {S, D, F} tatăl lui F = D tatăl lui D = S

{G, E, C, A, S} -> {S, A, C, E, G}

{T, E, C, A, S} -> {S, A, C, E, T}

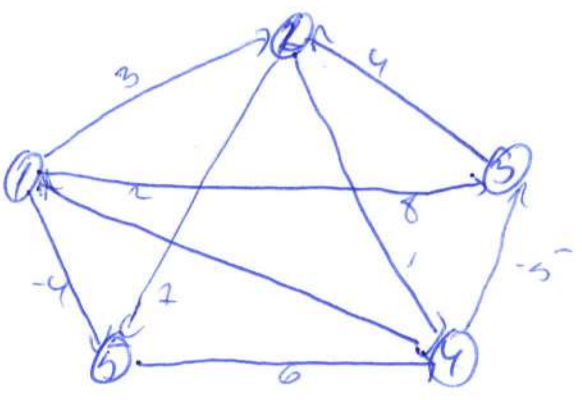
S -> A : {S, A}   S -> E : {S, A, C, E}

S -> B : {S; B}   S -> F : {S, D, F}

S -> C : {S, A, C}   S -> G : {S, A, C, E, G}

S -> D : {S, D}   S -> T : {S, A, C, E, T}

3.



	1	2	3	4	5
d	0	~~36~~ 3	-3	~~2~~ 4	~~2~~ 4
tată	0	~~0~~ 1	~~0~~ 4	~~0~~ 5	~~0~~ 1
count	~~0~~ 1	~~0~~ 1	~~0~~ 2	~~0~~ 1	~~0~~ 1

Q 1 2 5 4 3

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3) Vecinii lui 1: 2, 3, 5

$d_2 > d_1 + c_{12}$? $\cancel{\infty} > 0+3$? A

$d_3 > d_1 + c_{13}$? $\cancel{\infty} > 0+8$? A

$d_5 > d_1 + c_{15}$? $\cancel{\infty} > 0+(-4)$? A

Vecinii lui 2: 4, 5

$d_4 > d_2 + c_{24}$? $\cancel{\infty} > 3+1$? A

$d_5 > d_2 + c_{25}$? $-4 > 3+7$? F

Vecinii lui 3: 2

$d_2 > d_3 + c_{23}$? $3 > 8+4$? F

Vecinii 5: 4 (count 4=1)

$d_4 > d_5 + c_{54}$? $4 > -4+6$? A

Vecinii lui 4: 1, 3

$d_1 > d_4 + c_{14}$? $0 > 2+2$? F

$d_3 > d_4 + c_{43}$? $8 > 2+(-5)$? A

	1	2	3	4	5
d	0	~~3~~ 1	-3	2	-4
tată	0	~~1~~ 3	4	5	1
count	1	~~1~~ 2	2	1	1

Q 3 2

Vecinii lui 3: 2

$d_2 > d_3 + c_{32}$? $3 > -3+4$? A

Vecinii lui 2: 4, 5

$d_4 > d_2 + c_{24}$? $2 > 1+1$? F

$d_5 > d_2 + c_{52}$? $-4 > 1+7$? F Coada este goală, nu se mai poate extrage

1 -> 2: 1(-4+6-5+4)
1 -> 3: -3(-4+6-5)
1 -> 4: 2(-4+6)
1 -> 5: -4

1 -> 2 nodurile {1, 5, 4, 3, 2}
1 -> 3 nodurile {1, 5, 4, 3}
1 -> 4 nodurile {1, 5, 4}
1 -> 5 nodurile {1, 5}